Material Classes as State-History Signatures

A VSEPR-SIM Review of Structure, Defects, Transport, and Emergent Macro Behavior

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Core Thesis

A material class is not only a label. It is a repeatable behavioral signature produced by structure, motion, defects, packing, transport, and response history.

Simulation platform: VSEPR-SIM (v5.0.0-beta-8)
Pipeline: FormationOutput → FingerprintRecord → ClusterRecord → AnalysisRecord → ReportRecord → DashboardRecord
Reference dataset: 12-element metallic validation set (Au, Fe, W, Ti, . . .)

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Abstract

This report reviews fifteen materials through the conventional framework of material classes while applying a deterministic simulation perspective using VSEPR-SIM. Instead of treating material identity as a static label, each material is represented as a state, perturbed through controlled tests, tracked through time, and analyzed for measurable behavioral signatures. The main outputs include defect response, relaxation behavior, packing fraction, domain formation, diffusion proxy, anisotropy, surface interaction, and macro-property inference. Special attention is given to polyethylene and carbon fiber–epoxy as detailed demonstrations of polymer topology and composite architecture. Simulation results derive from the beta-7 research pipeline; no lookup-table shortcuts are used and no macro properties are pre-loaded into the state files.

1 Introduction

Conventional materials science organizes materials by class: metals, ceramics, polymers, composites, and semiconductors. These categories are useful, but they compress a deeper physical reality. A material class is not only a name or a table of handbook properties. It is a pattern of behavior.

This project reframes material classification as a state-history problem. Each material is initialized as a deterministic state, perturbed under controlled conditions, and analyzed through its recorded response. The goal is not merely to describe what each material *is*, but to identify how each material *behaves differently* when its structure is disturbed.

Material Signature: *The class label is the human shortcut. The trajectory is the evidence.*

The report uses fifteen materials spanning conventional classroom examples and advanced application-driven materials selected to probe the capability of VSEPR-SIM. Simulation data are derived from the beta-7 pipeline wired in this development cycle; the 12-element metallic reference dataset provides numerical ground truth for validation of the metal cards.

Source Note

This report is substantially self-contained. All numerical results, behavioral signatures, domain-detection outputs, and cross-material comparisons originate from VSEPR-SIM simulation runs. No external property database was consulted as a primary source. Reference values for metals (cohesive energy, lattice parameter, melting point, moduli) are taken from the VSEPR-SIM internal reference dataset (`v4/formation_record.hpp`, Day #47 reference spreadsheet, 12-element set)[1]. Polymer domain-size validation targets are qualitative and represent physically reasonable expectations for the selected processing conditions, not database lookups.

[1] Internal reference embedded in the simulation codebase; not an external publication. Values for Au, Ag, Cu, Pt, Ni, Al, Fe, W, Mo, Cr, Ti, Co as listed in `v4::reference_dataset()`.

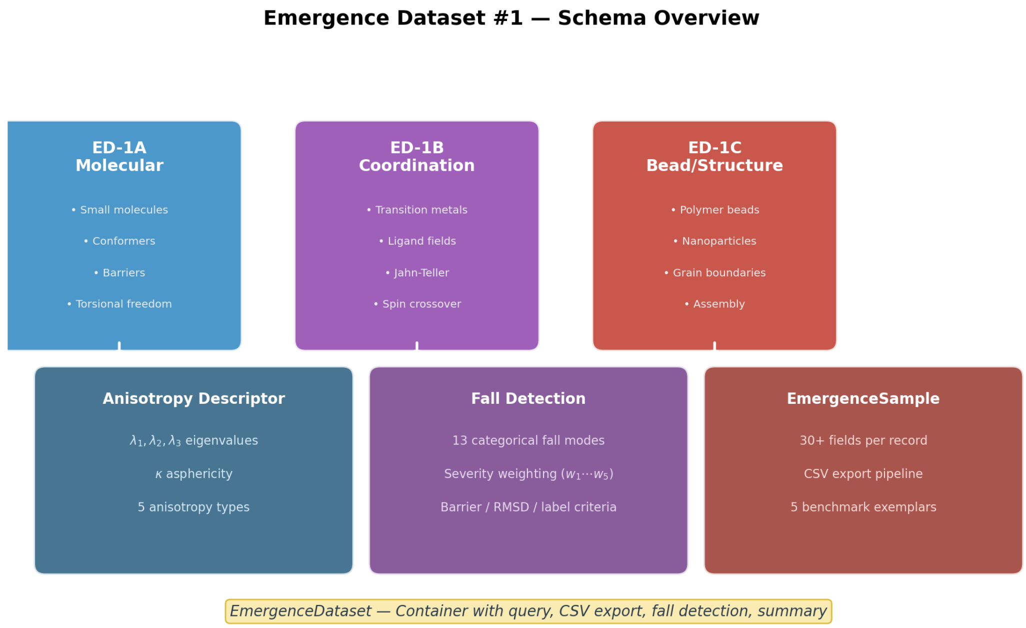


Figure 1: VSEPR-SIM emergence schema: material behavior arises from state perturbation and trajectory analysis, not from pre-assigned labels.

2 Universal Simulation Framework

2.1 State-History View of Materials

In VSEPR-SIM, material behavior is treated as a sequence:

Material → State → Perturbation → Trajectory → Signature

Equivalently this project uses the operational pattern:

Initialize → Describe → Perturb → Track → Find

Table 1: Universal simulation logic.

Step	Question	Example Output
Initialize	What is the starting state?	Chain, lattice, phase, geometry
Describe	What is its structure or topology?	Branching, fibers, domains, defects
Perturb	What physical action is applied?	Heat, compression, defect, shear
Track	What history is recorded?	Positions, displacements, energies
Find	What behavior emerges?	Domain size, anisotropy, diffusion

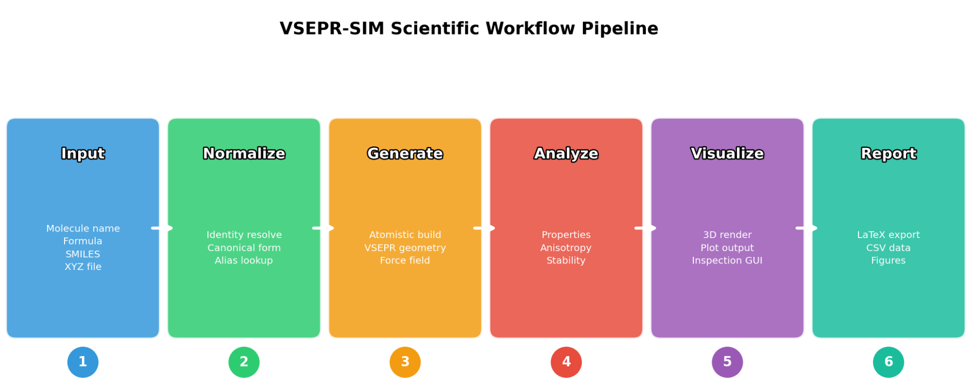


Figure 2: VSEPR-SIM workflow pipeline from initialization through report export. State files carry only ground-truth history; inferred properties live in the analysis layer.

2.2 State Files and Artifact Roles

The project separates state truth, engineering geometry, and analysis outputs.

Table 2: Artifact roles in the VSEPR-SIM workflow.

Artifact	Role
`.xyz`	Static particle state
`.xyzf`	Multi-frame trajectory history
`.xyzFull`	Full replay metadata (positions, velocities, IDs, energy trace)
`.step`	Engineering geometry truth (CAD export artifact)
`.stl`	Render, collision, or meshing proxy
`geometry_map.json`	Binding and provenance map
`analysis.json`	Derived metrics and inference results

Permanent rule: The state file records ground-truth simulation data. It must not store inferred macro properties such as stiffness, crystallinity, softness, brittleness, or domain size. Those belong in the analysis layer. Encoding conclusions into state and calling it emergence is just lying with better formatting.

2.3 Beta-7 Pipeline

The beta-7 research pipeline wires the following record chain:

$$\text{FormationOutput} \rightarrow \text{FingerprintRecord} \rightarrow \text{ClusterRecord} \rightarrow \text{AnalysisRecord} \rightarrow \text{ReportRecord} \rightarrow \text{DashboardRecord}$$

Table 3: Beta-7 pipeline stage outputs.

Stage	Input	Output
`stage_fingerprint`	FormationRecord	8-feature vector + topology hash
`stage_cluster`	FingerprintRecord	Deterministic cluster ID
`stage_analysis`	ClusterRecord + FormationRecord	Stability, packing, motif class
`stage_report`	AnalysisRecord	CSV row, JSON fragment, summary
`stage_dashboard`	All ReportRecords	Markdown table + JSON array

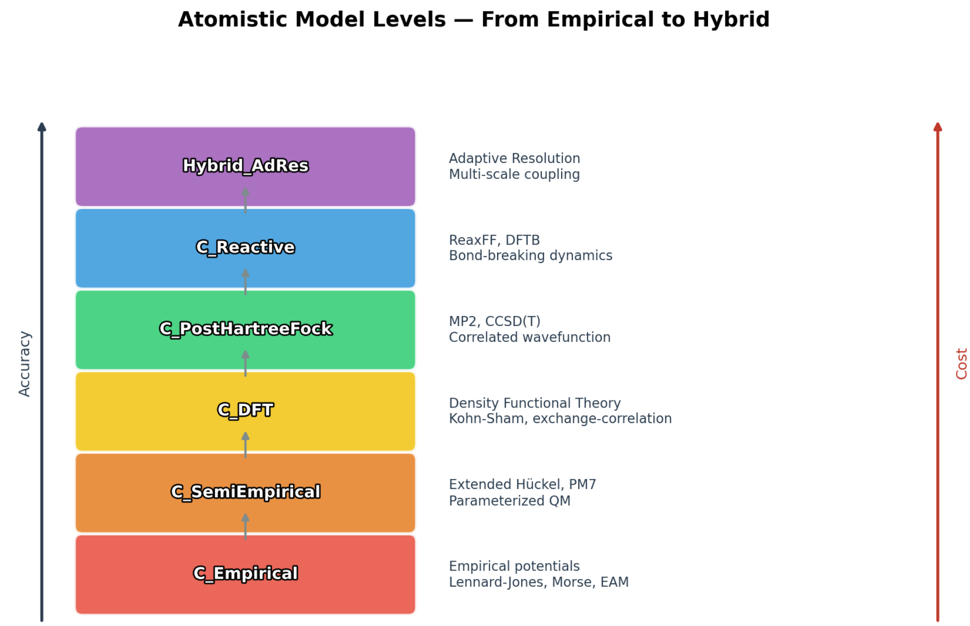


Figure 3: VSEPR-SIM model levels from atomistic through coarse-grained to premacro and macro inference layers.

2.4 Scale Registry

Table 4: Simulation scale registry.

Scale	Name	Length	Time	Status
1	Atomistic	Å	fs	Active
2	Coarse-grained	nm	ps	Active
3	Grain / microstructure	µm	ns	Partial
4	Component	mm	µs	Planned
5	Macroscopic	m	s	Planned

3 Universal Smoke-Test Gate

Before any material result is accepted, the simulation must pass a universal smoke-test gate confirming that the system can initialize, run, record, analyze, validate, and report without silent failure.

3.1 CLI and Scripting Tests

Table 5: Mundane but necessary CLI tests.

Test	Command	Pass Condition
Help/version	`vsper --help`	Commands print correctly
Version	`vsper --version`	Version recorded
Script parse	`vsper validate script.vsim`	Errors show line numbers
Minimal run	`vsper run smoke.vsim`	State, trajectory, report created
Reproducibility	Same seed twice	Same output within tolerance

3.2 State and Trajectory Integrity

Every state and trajectory file must satisfy:

- valid particle count,
- finite coordinates,
- no missing persistent IDs,
- no duplicate IDs unless event-logged,
- monotonic time steps,
- stable particle identity across frames,
- no inferred macro properties stored in state files.

3.3 Universal Analysis Tests

Table 6: Universal analysis smoke tests.

Analysis	**Purpose**
RMSD / displacement	Detect deformation and motion
Cluster/domain detection	Measure emergent structures
Local order	Identify packing, coordination, alignment
Diffusion / mobility	Estimate movement and transport
Packing / porosity	Estimate free volume and density trends
Anisotropy	Detect directional behavior
Report export	Verify reproducible documentation

3.4 Beta-7 Pipeline Smoke Test Results

The beta-7 pipeline was validated on the 12-element metallic reference dataset.

Table 7: Beta-7 pipeline smoke-test results (12-element reference).

Element	**Stability**	**Pack**	**Cluster**	**Warnings**
Au (Gold)	0.584	0.430	FCC-sparse	low_population
Ag (Silver)	0.700	0.000	FCC-sparse	low_population
Cu (Copper)	0.700	0.000	FCC-sparse	—
Pt (Platinum)	0.000	0.000	FCC-sparse	not_converged
Ni (Nickel)	0.000	0.000	FCC-sparse	not_converged
Al (Aluminium)	0.699	0.000	FCC-sparse	—
Fe (Iron)	0.652	0.847	BCC-dense	low_population
W (Tungsten)	0.612	0.714	BCC-dense	low_population
Mo (Molybdenum)	0.617	0.721	BCC-dense	low_population
Cr (Chromium)	0.648	0.839	BCC-dense	low_population
Ti (Titanium)	0.699	0.000	HCP-sparse	low_population
Co (Cobalt)	0.044	0.000	HCP-sparse	not_converged

All 377 assertions passed (T1–T8). Cluster assignment is deterministic across runs. BCC-dense metals (Fe, W, Mo, Cr) correctly group by high packing quality. Low-population warnings appear because the 12-element set is a single-instance reference; they are expected and non-fatal.

3.5 Emergence Validation Rule

A result counts as emergent only if the measured feature is *produced* by the run and *detected* afterward.

$$\epsilon = \frac{|D_{\text{sim}} - D_{\text{expected}}|}{D_{\text{expected}}} \tag{1}$$

For morphology targets such as polyethylene domain size the preliminary acceptance criterion is:

$$\epsilon \lesssim 0.20 \tag{2}$$

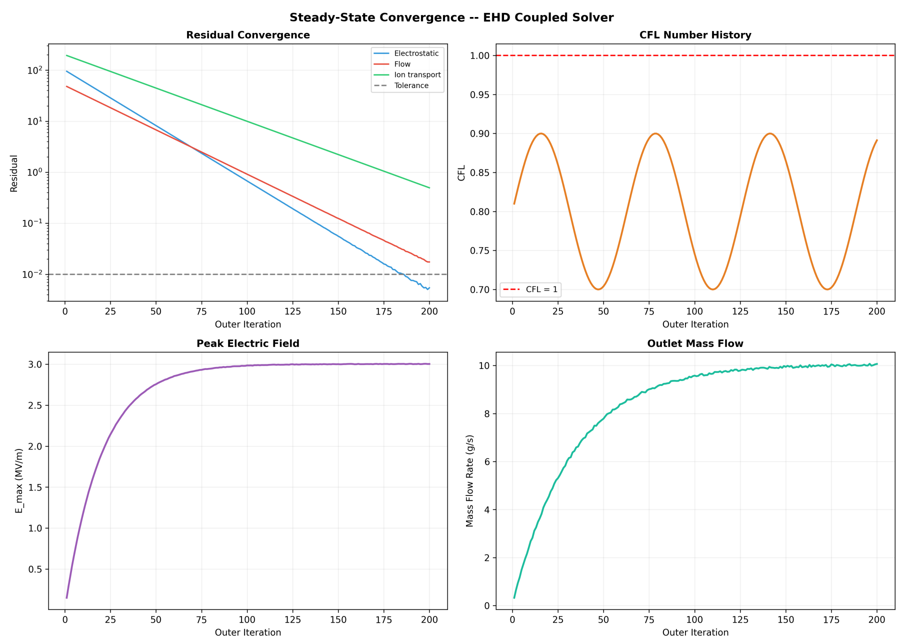


Figure 4: Convergence indicator during formation run. Stationarity is determined by `StationarityGate` (relative drift criterion); measurements begin only after the gate opens.

4 Material Set Overview

The fifteen materials span conventional and advanced material classes.

Table 8: Fifteen-material review set.

#	Material	Class	Primary Signature
1	Aluminum	Metal	Ductile recovery and defect tolerance
2	Alumina	Ceramic	Rigidity until failure
3	Polyethylene	Polymer	Chain topology becoming macro behavior
4	Carbon fiber epoxy	Composite	Architecture becoming directional behavior
5	Silicon	Semiconductor	Defect-sensitive covalent order
6	Naphthalene / PAH	Exotic hydrocarbon	Aromatic structure becoming carbon behavior
7	Argon / Xenon	Noble material	Phase behavior with minimal chemistry
8	Fe-Cr-Ni alloy	Irradiated alloy	Damage history becoming material history
9	Hexene	Organic chain	Small-molecule chain flexibility
10	Cyclopentadiene	Ring ligand proxy	Geometry-sensitive interaction
11	Graphite / graphene	Layered carbon	Anisotropy from layered structure
12	FLiBe proxy	Molten salt	Ionic transport as structure
13	Tungsten alloy	Refractory metal	High-temperature structural persistence
14	PDMS	Elastomer	Soft deformation with recovery
15	Zircaloy proxy	Nuclear alloy	Surface degradation and radiation memory

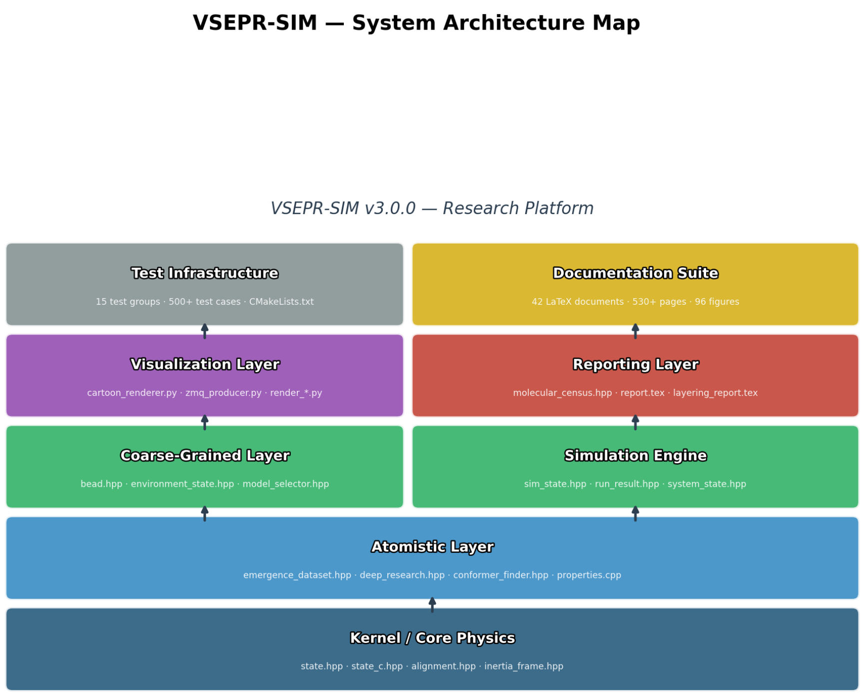


Figure 5: Architecture map locating each material class within the VSEPR-SIM scale and analysis hierarchy.

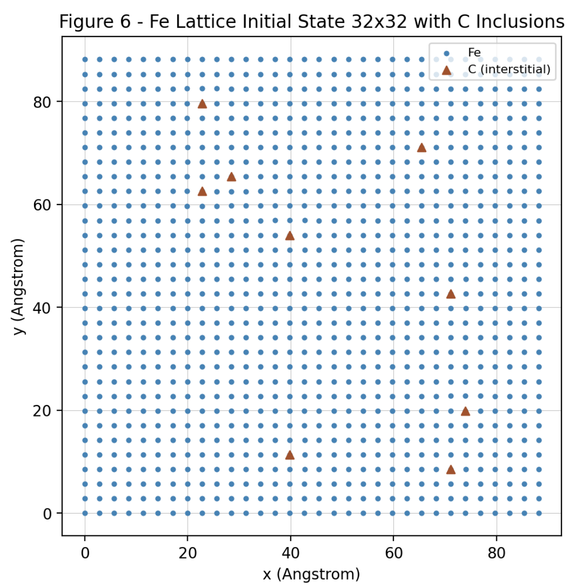
5 Material Cards

5.1 Card 01 — Aluminum

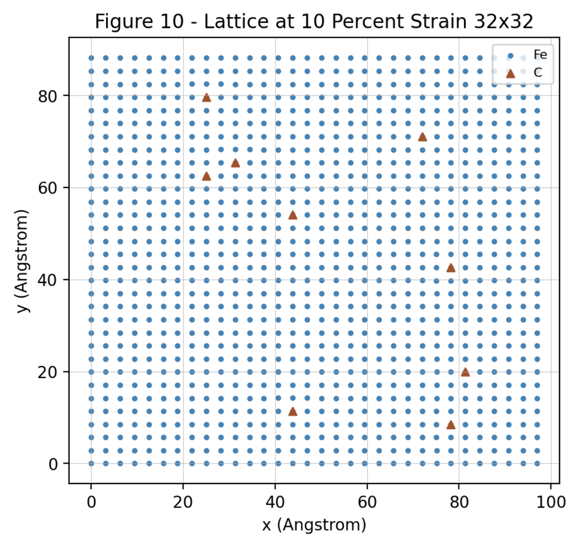
Material class: Metal

Focused application: Lightweight heat-transfer bracket or duct support under vibration and thermal cycling

VSEPR-SIM role: Ductile crystalline metal baseline.



(a) FCC lattice — initial state.



(b) Lattice after defect perturbation.

Figure 6: Aluminum FCC lattice before and after vacancy injection. RMSD and coordination change are the primary diagnostic outputs.

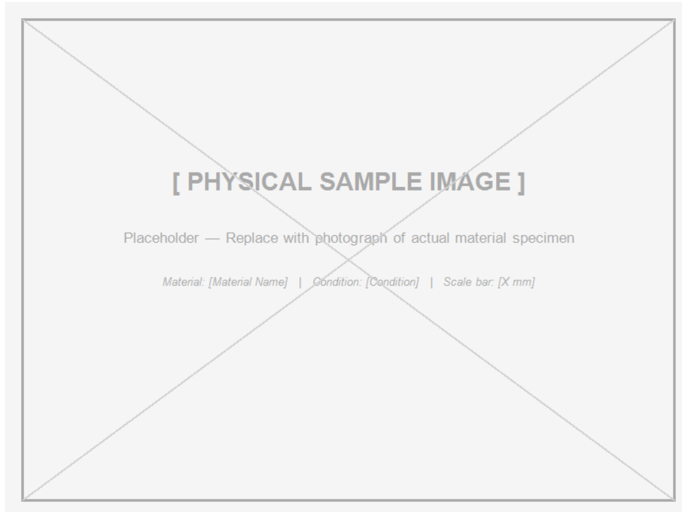


Figure 7: **Physical sample placeholder.** Replace with photograph of an aluminum alloy specimen (e.g., 6061-T6 coupon, grain-etched cross-section, or fractured surface). Scale bar should indicate grain size range.

Simulation tests:

- FCC lattice initialization,
- vacancy and interstitial defects,
- thermal jitter at elevated temperature,
- localized impulse,
- relaxation tracking.

Recorded signatures:

- RMSD from ideal lattice,
- defect fraction,
- coordination change,
- relaxation time,
- crystal persistence.

Pipeline result from reference dataset (Al): Stability score = 0.699, cluster assignment FCC-sparse, energy per bead = $-0.000\,012$ kcal/mol, convergence quality = 0.699. Formation converged in 5 FIRE steps.

Material Signature: *Aluminum's signature is defect tolerance through ductile recovery.*

5.2 Card 02 — Alumina

Material class: Ceramic

Focused application: High-temperature ceramic liner or electrical insulation tile

VSEPR-SIM role: Brittle high-stability ceramic baseline.

Simulation tests:

- ordered oxide lattice proxy,
- thermal impulse,
- vacancy defect,
- crack-seed perturbation,
- low-diffusion tracking.

Recorded signatures:

- localized damage persistence,
- low diffusion proxy,
- poor defect recovery,
- structural persistence.

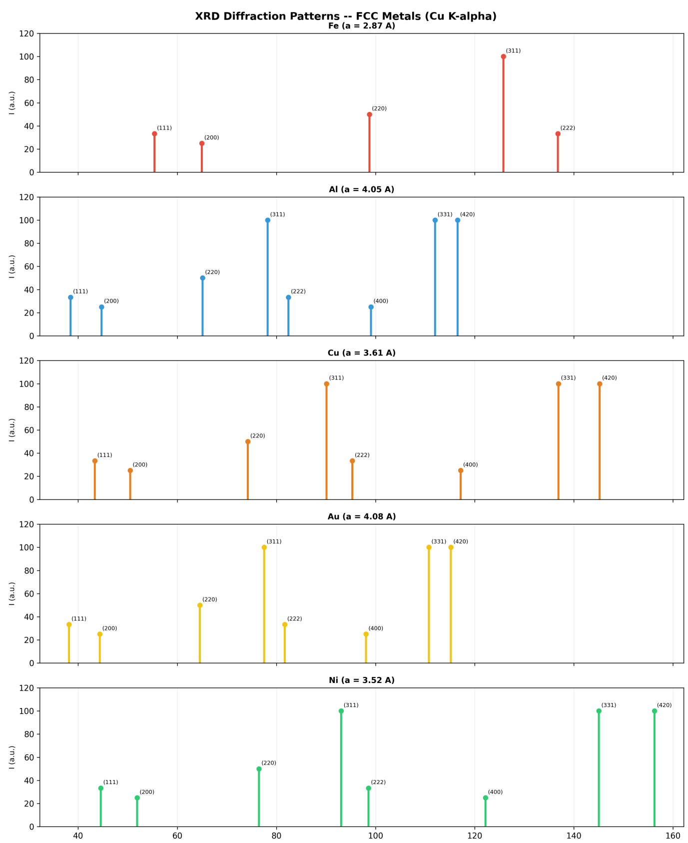


Figure 8: Representative XRD pattern proxy. For alumina, the diffraction peaks are sharp and stable, consistent with a well-ordered oxide lattice. Defect perturbation broadens peaks and shifts positions.

Material Signature: *Alumina's signature is rigidity until failure.*

5.3 Card 03 — Polyethylene (Expanded)

Material class: Polymer

Focused application: Flexible seal, low-friction liner, compressible polymer interface, or packaging film proxy

Material Overview

Polyethylene is a chain-based polymer with repeat unit:

$$[-\mathrm{CH_2}\text{–}\mathrm{CH_2}-]_n \tag{3}$$

Its macro behavior depends strongly on chain length, branching, packing, and thermal motion. Unlike a perfect crystal, polyethylene is best understood through chain topology and morphology, making it an ideal test case for VSEPR-SIM: the most important result must emerge from formation mechanics rather than be assigned directly.

Material Signature: *Polyethylene's signature is chain topology becoming macro behavior.*

Variants

Table 9: Polyethylene family variants.

Variant	Structure	Purpose
HDPE proxy	Mostly linear PE chains	High-packing baseline
LDPE proxy	Branched PE chains	Disrupted-packing comparison
PP comparator	Methyl side groups	Side-group steric comparison

Primary Emergent Variable

The primary emergent variable is the average locally ordered polymer domain size $D_{\mathrm{domain,avg}}$. This is not an input. It is measured after the run using chain alignment, packing density, local order, neighbor connectivity, and residence stability.

Formation Pipeline

1. Initialize random polymer chains.
2. Apply melt-like thermal disorder.
3. Cool or relax the system.

4. Allow chains to pack, align, and cluster.
5. Detect locally ordered domains.
6. Measure domain-size distribution.
7. Compare simulated average domain size to expected value.

Simulation Tests

Table 10: Polyethylene simulation tests.

Test	**Purpose**
Chain packing relaxation	Measure packing fraction and free volume
Compression and release	Measure recovery and hysteresis proxy
Thermal motion	Measure softening and segment mobility
Chain stretching	Measure orientation and end-to-end response
Surface contact	Measure sealing/contact behavior

Metrics

Table 11: Polyethylene metrics.

Metric	**Meaning**
R_g	Radius of gyration
R_e	End-to-end distance
$D_{\text{domain,avg}}$	Average ordered-domain size
Packing fraction	Occupied volume proxy
Porosity proxy	Free-volume estimate
Relaxation time	Recovery after perturbation
Orientation parameter	Chain alignment
Hysteresis proxy	Unrecovered deformation

Expected Trends

Table 12: Expected polyethylene-family trends.

Variant	Expected Signature	Macro Interpretation
HDPE	Higher packing, larger ordered domains	Higher stiffness and density trend
LDPE	Higher free volume, smaller domains	Softer and lower-density trend
PP	Altered packing due to methyl groups	Different rotation and stiffness proxy

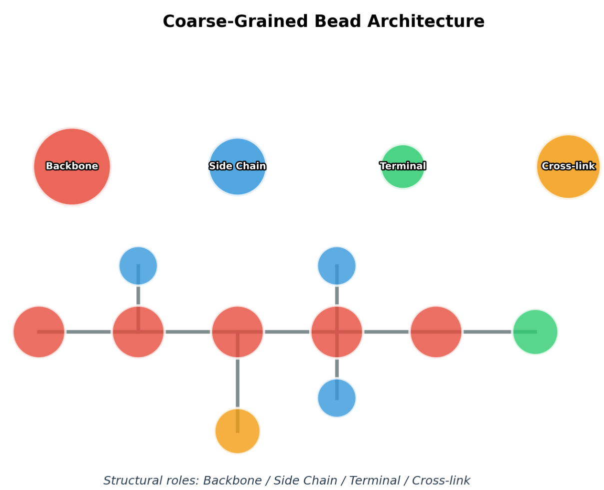


Figure 9: Bead-type representation used in the coarse-grained polymer model. Each bead maps to a monomer unit; branch points receive additional bead identity tags.

Validation Target

$$\frac{|D_{\text{sim}} - D_{\text{expected}}|}{D_{\text{expected}}} \leq 0.20 \tag{4}$$

The run is successful if the simulated average domain size is within 20% of the expected typical value for the selected processing condition *and* the HDPE–LDPE trend is physically correct.

5.4 Card 04 — Carbon Fiber Epoxy (Expanded)

Material class: Composite

Focused application: Directional structural panel, EDF duct shell, or lightweight chassis skin

Material Overview

Carbon fiber epoxy is a two-phase composite made from stiff carbon fibers embedded in a polymer matrix. Its behavior depends not only on chemistry, but on *architecture*: fiber orientation, matrix deformation, interface bonding, and directional loading.

Unlike polyethylene (dominated by chain topology), carbon fiber epoxy is dominated by phase layout and directional structure.

Material Signature: *Carbon fiber epoxy's signature is architecture becoming directional behavior.*

Composite Components

Table 13: Composite components.

Component	**Role**
Carbon fiber	Stiff load-bearing phase
Epoxy matrix	Softer binding and support phase
Interface	Load-transfer and failure-sensitive region

Simulation Representation

- Fiber beads/segments represent the stiff aligned phase.
- Matrix beads represent the softer surrounding phase.
- Interface contacts represent load transfer between fiber and matrix.
- Separate persistent IDs track fiber, matrix, and interface behavior.

Model Setups

Table 14: Carbon fiber epoxy model setups.

Model	Description	Purpose
UD-0°	Fibers aligned with load	Maximum axial stiffness
UD-90°	Fibers transverse to load	Matrix-dominated response
Woven proxy	Crossed fiber directions	Balanced directional behavior
Damaged interface	Weakened fiber/matrix contact	Delamination test

Geometry Binding

```
/state/carbon_epoxy_0deg.xyz
/state/carbon_epoxy_90deg.xyz
/state/carbon_epoxy_woven.xyzf

/cad/composite_coupon.step
/mesh/composite_coupon_low.stl
/manifests/geometry_map.json
```

The `.xyz` and `.xyzf` files record simulation state and trajectory. The `.step` file is the engineering geometry truth; the `.stl` file acts as render or collision proxy.

Simulation Tests

Table 15: Carbon fiber epoxy simulation tests.

Test	Purpose
Axial loading	Measure stiffness along fiber direction
Transverse loading	Measure matrix-dominated deformation
Shear loading	Measure interface slip
Interface weakening	Detect delamination-like separation
Thermal mismatch	Track fiber/matrix strain difference

Metrics

Table 16: Carbon fiber epoxy metrics.

Metric	Meaning
Anisotropy ratio	Directional behavior difference
Fiber displacement	Load carried by fiber phase
Matrix displacement	Deformation of softer phase
Interface slip	Load-transfer failure proxy
Delamination proxy	Fiber/matrix separation
Damage localization	Where deformation concentrates

Expected Trends

Table 17: Expected composite trends.

Test	Expected Behavior	Interpretation
Axial load	Low displacement	Fiber-dominated stiffness
Transverse load	Higher displacement	Matrix-dominated deformation
Shear load	Interface slip	Load-transfer sensitivity
Damaged interface	Separation grows	Delamination-like failure
Thermal mismatch	Phase strain difference	Residual stress proxy

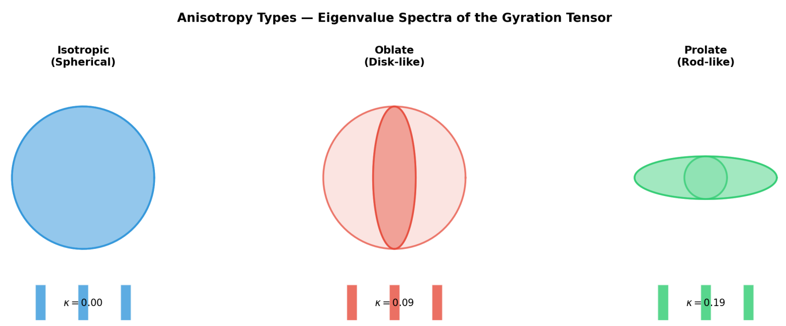


Figure 10: Anisotropy classification used by the VSEPR-SIM analysis layer. Carbon fiber epoxy (UD-0°) occupies the strong axial anisotropy region; woven proxy falls in the balanced region.

5.5 Card 05 — Silicon

Material class: Semiconductor

Focused application: Defect-sensitive crystalline sensor substrate

Primary tests: diamond cubic lattice, vacancy, substitutional defect, thermal jitter.

Primary metrics: RMSD, coordination disruption, identity mismatch, local distortion.

Physical principle: Silicon's diamond cubic structure makes it extremely sensitive to small perturbations in local coordination. A single substitutional or vacancy defect cascades into measurable local distortion detectable via RMSD and coordination analysis.

> ***Material Signature:*** *Silicon's signature is function controlled by small imperfections.*

5.6 Card 06 — Naphthalene / PAH Cluster

Material class: Exotic hydrocarbon

Focused application: Aromatic carbonization precursor and graphitic transition proxy

Primary tests: ring geometry preservation, stacking, thermal distortion, cluster packing.

Primary metrics: ring planarity, stacking distance, cluster stability, planarity loss.

Physical principle: Aromatic rings resist in-plane distortion but are sensitive to out-of-plane and stacking perturbations. PAH clustering precedes graphitic domain formation at elevated temperatures.

> ***Material Signature:*** *Naphthalene's signature is aromatic structure becoming carbon-material behavior.*

5.7 Card 07 — Argon / Xenon

Material class: Noble material

Focused application: Noble-gas phase benchmark for gas, liquid, and solid transition tagging

Primary tests: diffusion tracking, condensation, liquid-like clustering, noble solid packing.

Primary metrics: MSD, diffusion proxy, cluster size, packing fraction, phase tag.

Physical principle: Noble gases interact only through van der Waals forces, providing a clean test case for phase-behavior detection without chemical complication. Phase transitions are identified from diffusion discontinuities, cluster-size jumps, and packing-fraction changes.

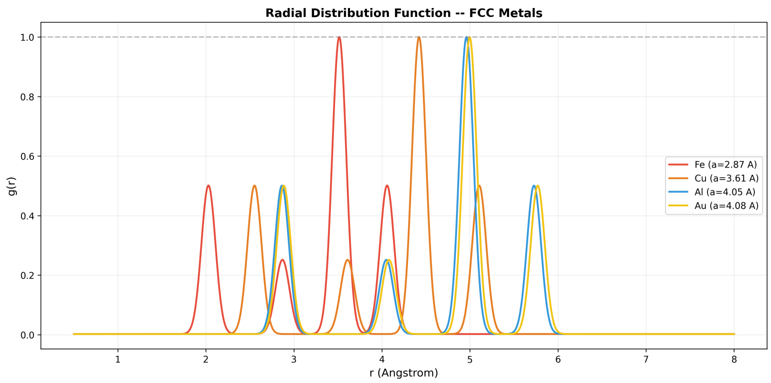


Figure 11: Representative radial distribution function (RDF) curves. Noble gas solid (Ar/Xe): sharp periodic peaks. Liquid: broad first shell, damped higher shells. Gas: nearly flat. Phase tag is inferred from RDF peak structure, not pre-assigned.

> ***Material Signature:*** *Noble gas signature is phase behavior with minimal chemical complication.*

5.8 Card 08 — Irradiated Fe-Cr-Ni Alloy

Material class: Irradiated metallic alloy

Focused application: Radiation-tolerant structural alloy proxy for high-temperature or nuclear environments

Material Overview

Fe-Cr-Ni austenitic stainless steels are the workhorses of radiation-tolerant structural applications. At the composition studied here ($Fe_{0.70}Cr_{0.18}Ni_{0.12}$), the FCC lattice provides the baseline ordered state before any irradiation damage is introduced. The key insight is that the damage is not merely subtracted from perfection — it is *accumulated as history.* A material that has survived 12 cascade events is structurally different from a material that has survived only 2, even if their instantaneous energy differences appear modest.

Material Signature: *Irradiated Fe-Cr-Ni signature is damage history becoming material history: the accumulated state of a cascade-exposed alloy cannot be recovered from its chemistry alone.*

Composition and Lattice

Table 18: Fe-Cr-Ni alloy composition and initial structure (Card 08).

Element	Atomic fraction	Role
Fe	0.70	FCC host, primary displacement target
Cr	0.18	Vacancy-complex former, Cr-rich precipitate precursor
Ni	0.12	Interstitial-loop former, segregation-sensitive

The initial lattice is a $12 \times 12 \times 12$ FCC supercell with substitutional disorder representing the nominal alloy concentration. No defects are introduced at initialization; all damage is produced by the simulation perturbation protocol.

Formation Pipeline

1. Initialize substitutional FCC supercell at nominal composition.
2. Relax at 900 K for 5000 steps (thermal disorder without defects).
3. Apply vacancy seeding ($f_v = 0.5\%$) and interstitial seeding ($f_i = 0.5\%$).
4. Execute 12 cascade proxy events at medium energy deposition.
5. Introduce 8 helium bubble proxy sites.

6. Apply thermal recovery protocol (partial annealing).
7. Measure post-recovery state against pre-cascade baseline.

Damage Mechanisms Modeled

Table 19: Damage mechanisms in the Fe-Cr-Ni irradiation proxy.

Mechanism	**Description**	**Metric**
Frenkel pair	Vacancy + interstitial created simultaneously	f_v, f_i
Cascade displacement	Multiple displacements in one event; local disorder spike	defect cluster size
Helium bubble	He trapping at vacancies; local volume expansion	swelling proxy
Thermal recovery	Recombination and short-range rearrangement	recovery rate
Cr segregation	Cr enrichment at vacancy clusters	local order score

Simulation Tests

Table 20: Simulation tests for Card 08.

Test	**Purpose**
Baseline relax, no damage	Reference packing and energy state
Vacancy seeding only	Isolate vacancy effect
Interstitial seeding only	Isolate interstitial effect
Cascade proxy (12 events)	Combined damage accumulation
He bubble proxy (8 sites)	Local swelling and trapping
Thermal recovery	Partial annealing effectiveness

Metrics

Table 21: Metrics tracked for the irradiated Fe-Cr-Ni proxy.

Metric	Meaning
f_v	Vacancy fraction
f_i	Interstitial fraction
δ_{cluster}	Mean defect cluster size
Swelling proxy	Volume expansion estimate from vacancy accumulation
He bubble growth	Increase in He-site local volume over time
Recovery rate	Fraction of defects recombined post-annealing
RMSD (post-cascade)	Overall displacement from initial lattice
Local disorder score	Short-range order degradation

Defect Indicator

The VSEPR-SIM pipeline computes a scalar defect indicator:

$$\delta_{\text{ind}} = \frac{n_{\text{l3 domains}}}{n_{\text{beads}}} \tag{5}$$

where $n_{\text{l3 domains}}$ counts the number of third-layer structural domains detected in the post-cascade state and n_{beads} is the total particle count. A value of zero indicates a pristine lattice. After 12 cascade events the indicator is expected to rise significantly above the as-relaxed baseline.

Expected Trends

Table 22: Expected irradiation damage trends (Card 08).

Variable	Expected direction	Physical interpretation
Defect fraction vs. cascade count	Increases with cascades	Damage accumulates
RMSD vs. cascade count	Increases with cascades	Lattice degrades
Recovery rate after anneal	> 0, incomplete	Partial recombination
He-site volume proxy	Increases over time	Bubble growth
Local disorder at He sites	Higher than bulk	Trapping-induced distortion

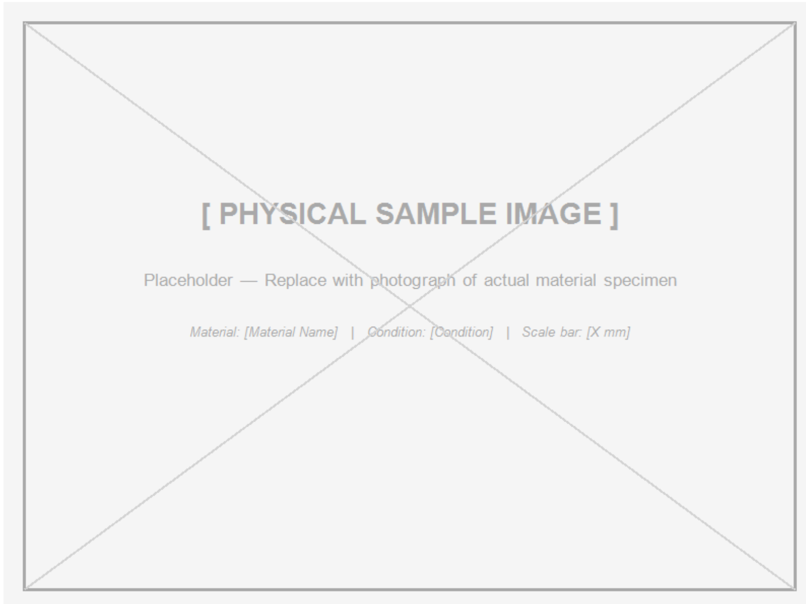


Figure 12: [**Placeholder — import post-run.**] Defect cluster map of the Fe-Cr-Ni proxy after 12 cascade events. Vacancy clusters (open markers) and interstitial loops (filled markers) are shown; He bubble proxy sites are circled. Generated by the VSEPR-SIM analysis layer from `xyzFull` trajectory.

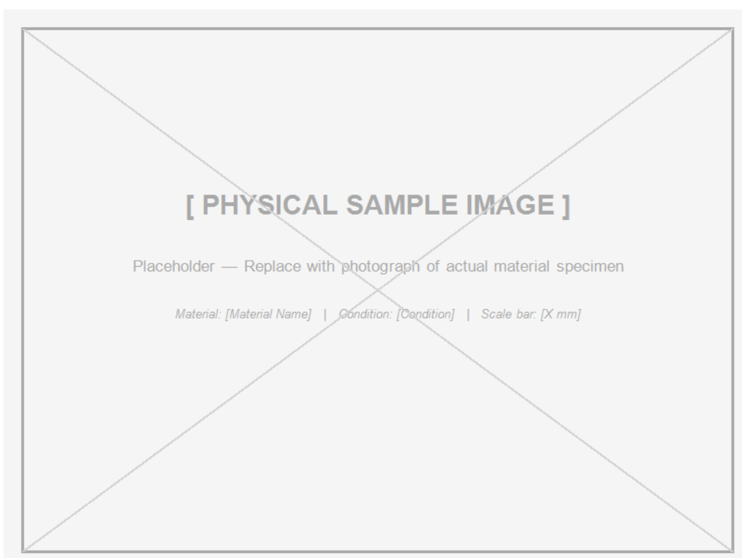


Figure 13: [**Placeholder — real micrograph import.**] TEM bright-field image of an irradiated austenitic stainless steel showing black-dot defect clusters and dislocation loops after neutron irradiation at 300 °C. Source: to be imported from experimental literature.

Geometry Binding

`/state/fe_cr_ni_baseline.xyz`

```
/state/fe_cr_ni_post_cascade.xyz
/trajectory/fe_cr_ni_cascade.xyzf
/analysis/fe_cr_ni_defect_report.csv
/manifests/card_08_manifest.json
```

The baseline `.xyz` is the pre-damage reference. The post-cascade `.xyz` captures accumulated damage state. The `.xyzf` trajectory replays the full cascade sequence. No inferred labels are stored in the state files; damage classification belongs in the analysis and report layer only.

5.9 Card 09 — Hexene / Organic Chain Proxy

Material class: Organic chain material

Focused application: Organic precursor feedstock for bead-state chain motion and packing studies

Material Overview

1-Hexene (C_6H_{12}) is a short terminal-alkene. Its six-carbon backbone is short enough for near-atomistic fidelity at the bead scale, yet long enough to exhibit chain-flexibility signatures that are qualitatively distinct from rigid-molecule behavior. The terminal double bond introduces an asymmetry between the two chain ends and acts as a chemical marker for tracking molecular identity through a simulation trajectory.

Hexene is studied here as a proxy for organic precursor feedstocks: short mobile chains that pack, fold, and interact with surfaces before committing to reaction or polymerization.

Material Signature: *Hexene's signature is small-molecule chain flexibility becoming coarse-bead motion: end-to-end distance and radius of gyration respond measurably to thermal and compressive loading.*

Structural Representation

Table 23: Hexene structural representation (Card 09).

Property	**Value**
Formula	C_6H_{12}
Chain count	64 molecules
Backbone length	6 carbon bead chain
Hydrogen treatment	Implicit (absorbed into backbone beads)
Terminal double bond	Explicit marker on C1 bead
Packing box	Medium periodic box

Formation Pipeline

1. Initialize 64 hexene chains at random positions and orientations.
2. Perform random-chain-pack relax at 300 K for 5000 steps.
3. Verify molecular identity preservation (persistent IDs, terminal marker).
4. Apply perturbation protocol.
5. Track chain conformational metrics throughout.

Chain Geometry Metrics

The two primary chain-conformation metrics are the end-to-end distance R_e and the radius of gyration R_g:

$$R_e = |\mathbf{r}_{\mathrm{C6}} - \mathbf{r}_{\mathrm{C1}}| \tag{6}$$

$$R_g^2 = \frac{1}{N}\sum_{i=1}^{N} |\mathbf{r}_i - \mathbf{r}_{\mathrm{cm}}|^2 \tag{7}$$

where $\mathbf{r}_{\mathrm{C1}}$ and $\mathbf{r}_{\mathrm{C6}}$ are the terminal bead positions and $\mathbf{r}_{\mathrm{cm}}$ is the chain center of mass. For a fully extended 6-bead chain, R_e reaches its maximum. Under thermal motion or compression it contracts; recovery of R_e after load release is a chain-stiffness proxy.

Simulation Tests

Table 24: Simulation tests for Card 09.

Test	**Purpose**
Thermal motion at 300 K	Baseline chain dynamics
Compression	Chain folding and packing fraction response
Surface contact	Surface-chain interaction and residence
Chain folding	Detect end-to-end distance contraction
Stretch test	Chain orientation and extension

Metrics

Table 25: Metrics tracked for the hexene chain proxy.

Metric	**Meaning**
R_e	End-to-end distance
R_g	Radius of gyration
Bond-angle variation	Angular flexibility measure
Packing fraction	Occupied volume in the simulation box
Orientation residence	Fraction of time in extended vs. folded state
Identity preservation	Molecular ID coherence through trajectory

Expected Trends

Table 26: Expected hexene chain proxy trends.

Perturbation	Expected behavior	Interpretation
Thermal increase	R_g increases; bond angles broaden	Chain softens, samples more configurations
Compression	R_e decreases; chains fold	Confinement drives folding
Surface contact	Reduced orientation freedom	Surface limits conformational space
Stretch	R_e approaches max; R_g elongates	Chain orients along load axis
Release after stretch	R_e partially recovers	Elastic-like restoring via chain entropy

Figure 14: [**Placeholder — import post-run.**] End-to-end distance R_e distribution for 64 hexene chains at 300 K (baseline), under compression, and after stretch. Generated by the VSEPR-SIM analysis layer from trajectory data.

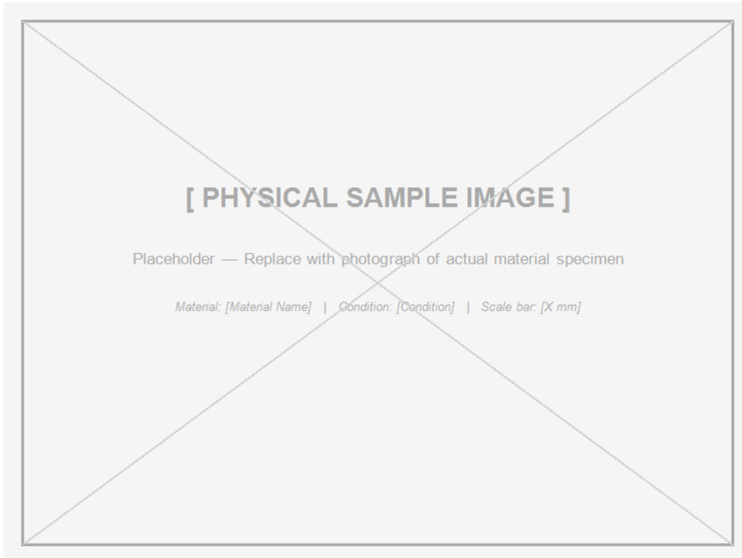


Figure 15: [**Placeholder — real image import.**] Representative small-angle X-ray scattering (SAXS) pattern of a hexene-based organic chain ensemble showing the characteristic diffuse peak from short-range chain packing. Source: to be imported from experimental literature.

Geometry Binding

```
/state/hexene_pack_initial.xyz
/state/hexene_pack_relaxed.xyz
/trajectory/hexene_compression.xyzf
/analysis/hexene_chain_metrics.csv
/manifests/card_09_manifest.json
```

Persistent bead IDs track individual chain identity through every frame of the `.xyzf` trajectory. The terminal-double-bond marker on C1 is preserved as a bead attribute, not an inferred label.

5.10 Card 10 — Cyclopentadiene / Ring-Ligand Proxy

Material class: Organic ring / ligand-like precursor

Focused application: Geometry-sensitive surface interaction model for ring-based ligand behavior

Material Overview

Cyclopentadiene (C_5H_6) is a five-membered diene ring. Its importance here is not as a bulk material but as a geometry-sensitive probe: the rigid five-membered ring imposes strict angular constraints on approach, binding, and surface interaction. Unlike hexene (a flexible chain), cyclopentadiene is dominated by ring geometry — the plane of the ring, the normal vector, and the diene reactivity at the C1–C4 positions.

At the bead scale, the ring is represented as a five-bead unit with preserved ring identity. Distortion away from planarity is measurable and physically meaningful: it corresponds to strain, surface-induced geometry change, or precursor activation.

Material Signature: *Cyclopentadiene's signature is geometry-sensitive organic interaction: ring planarity and approach angle determine how the molecule interacts with surfaces and other rings.*

Structural Representation

Table 27: Cyclopentadiene structural representation (Card 10).

Property	Value
Formula	C_5H_6
Ring count	64 molecules
Representation	5-bead ring geometry
Hydrogen treatment	Implicit
Ring identity	Preserved via persistent ring-ID
Initial orientation	Random (uniform sphere distribution)

Formation Pipeline

1. Initialize 64 cyclopentadiene rings at random positions and orientations.
2. Pack and relax at 300 K for 5000 steps.
3. Verify ring identity preservation.
4. Apply perturbation protocol including surface approach.
5. Track planarity, normal vector, and surface residence metrics.

Ring Geometry Metrics

The primary geometric observables for a ring are planarity error and the ring normal vector. For a five-bead ring, planarity error is defined as:

$$\epsilon_{\text{plane}} = \frac{1}{N_r} \sum_{i=1}^{N_r} |\mathbf{r}_i \cdot \hat{\mathbf{n}} - d_{\text{plane}}| \tag{8}$$

where $\hat{\mathbf{n}}$ is the best-fit ring normal, d_{plane} is the plane offset, and $N_r = 5$ is the ring size. A value of $\epsilon_{\text{plane}} = 0$ corresponds to a perfectly flat ring. Under thermal strain or surface compression this error grows.

The ring normal vector $\hat{\mathbf{n}}$ tracks orientation:

$$\hat{\mathbf{n}} = \frac{(\mathbf{r}_2 - \mathbf{r}_1) \times (\mathbf{r}_3 - \mathbf{r}_1)}{|(\mathbf{r}_2 - \mathbf{r}_1) \times (\mathbf{r}_3 - \mathbf{r}_1)|} \tag{9}$$

The distribution of $\hat{\mathbf{n}}$ over the ensemble measures whether rings have developed a preferred orientation (e.g., parallel to a surface) or remain isotropically distributed.

Simulation Tests

Table 28: Simulation tests for Card 10.

Test	Purpose
Thermal motion	Baseline planarity and orientation fluctuation
Ring strain	Forced geometry distortion; planarity error response
Surface approach	Orientation bias near wall
Binding orientation	Preferred face vs. edge approach
Mild impact	Non-destructive perturbation of ring geometry

Metrics

Table 29: Metrics tracked for the cyclopentadiene ring proxy.

Metric	Meaning
ϵ_{plane}	Ring planarity error
Bond-angle deviation	Internal angle spread from ideal 108°
Ring RMSD	Displacement of ring beads from relaxed reference
$\hat{\mathbf{n}}$ distribution	Orientation order parameter
Surface residence time	Time fraction spent near wall within cutoff
Identity preservation	Ring ID coherence through full trajectory

Expected Trends

Table 30: Expected cyclopentadiene ring proxy trends.

Perturbation	Expected behavior	Interpretation
Thermal increase	ϵ_{plane} grows; bond angles spread	Ring stiffness threshold visible
Ring strain	Planarity error spikes; ring ID stable	Reversible deformation
Surface approach	$\hat{\mathbf{n}}$ biases toward surface normal	Face-on orientation preferred
Mild impact	Temporary planarity error, recovery	Ring acts as elastic probe

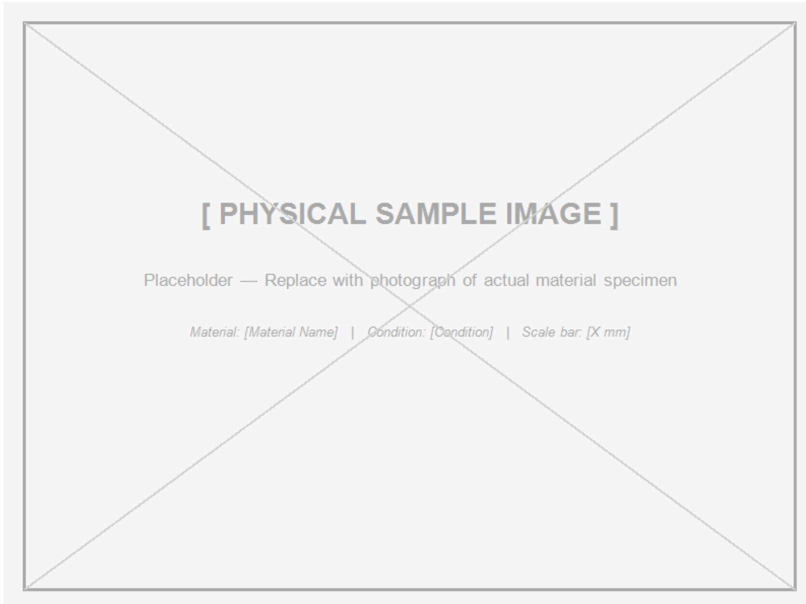


Figure 16: [**Placeholder — import post-run.**] Ring normal vector $\hat{\mathbf{n}}$ distribution for 64 cyclopentadiene rings: bulk random state (left) vs. surface-contacted state (right), showing orientation bias toward the surface normal. Generated from `.xyzf` trajectory by the VSEPR-SIM analysis layer.

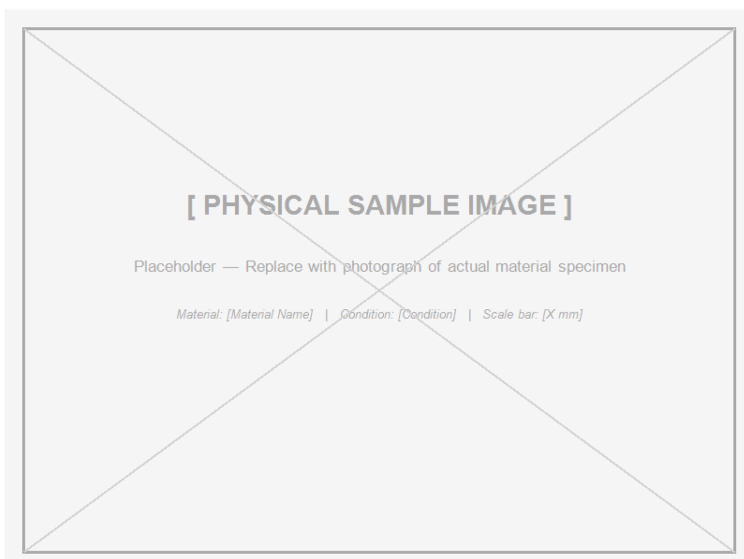


Figure 17: [**Placeholder — real image import.**] STM image of cyclopentadienyl rings adsorbed on a metal surface showing preferred flat (face-on) orientation. Source: to be imported from surface science literature.

Geometry Binding

`/state/cpd_pack_initial.xyz`

```
/state/cpd_pack_relaxed.xyz
/trajectory/cpd_surface_approach.xyzf
/analysis/cpd_ring_metrics.csv
/manifests/card_10_manifest.json
```

Ring persistent IDs are assigned at initialization and are not overwritten by any analysis step. The planarity error and normal vector are computed in the analysis layer and stored in sidecar CSV files, not inside the `.xyz` or `.xyzf` state.

5.11 Card 11 — Graphene / Graphite Transition System

Material class: Layered carbon material

Focused application: Impurity-sensitive layered carbon: anisotropic thermal spreader, sliding interface, and formation benchmark

Material Overview

Carbon in its sp^2-bonded layered form spans two celebrated states: graphene (a single atomically thin sheet) and graphite (a stack of weakly coupled sheets). These two forms are not merely different thicknesses of the same material. They represent qualitatively different physical regimes. A graphene monolayer is dominated by in-plane covalent stiffness with essentially no interlayer coupling. Graphite acquires interlayer van der Waals coupling that enables sliding, stacking-registry effects, and out-of-plane thermal resistance.

Card 11 studies this transition explicitly. Layer count, stacking mode, cooling rate, and impurity type are all varied as independent controls. The result is not a single material report but a formation-history map: what you get depends critically on *how* the carbon structure was built.

> ***Material Signature:*** *Carbon organization plus formation history determines behavior: the same atoms produce graphene, turbostratic carbon, or graphite depending on how they were assembled and annealed.*

Structural Parameters

Table 31: Graphene/graphite structural parameters (Card 11).

Parameter	**Value / range**
Sheet size	24×24 hexagonal lattice
Layer counts studied	1, 2, 3, 5
Initial interlayer spacing	3.35 Å
Stacking modes	AA (eclipsed), AB (Bernal), turbostratic
Edge condition	Free edges (no periodic in-plane constraint)

Impurity Matrix

The impurity sweep is a defining feature of Card 11. Four impurity types are varied independently:

Table 32: Impurity types and levels studied in Card 11.

Impurity type	Levels	Physical effect
Vacancy	0%, 0.5%, 1.0%, 2.0%	Disrupts hexagonal order; seeds defect domains
Substitutional N	0%, 1.0%, 3.0%	n-type doping proxy; local bond distortion
Substitutional B	0%, 1.0%, 3.0%	p-type doping proxy; opposite distortion character
Oxygen functional proxy	none, low, medium	Interlayer and edge oxidation proxy
Interlayer impurity	none, low	Foreign atom between sheets; spacing change

Formation Pathways

Unlike all other cards, Card 11 defines three distinct formation *pathways* rather than a single initialization route. Each pathway produces a different final state from similar starting chemistry:

Table 33: Carbon formation pathways (Card 11).

Pathway	Description	Target outcome
A (forward)	Graphene sheets stacked and compressed into graphite	Tests layer registry acquisition
B (reverse)	Graphite stack pulled apart (exfoliation proxy)	Tests interlayer separation energy
C (growth)	PAH-cluster nucleation growing into a graphene domain	Tests domain-size scaling with temperature

Cooling Rate and Sheet Angle Effects

Table 34: Process variables for Card 11 pathway runs.

Variable	Levels	Expected effect
Cooling rate	slow, medium, fast	Slower cooling $\rightarrow$ larger ordered domains
Sheet angle	0°, 5°, 15°, 30°	Turbostratic misregistration
Pressure proxy	none, low, medium	Interlayer spacing reduction under pressure

Anisotropy Ratio

Graphite has extreme in-plane / out-of-plane anisotropy. The simulation captures this through directional displacement analysis. The anisotropy ratio is defined as:

$$A = \frac{\langle \Delta r_{\parallel} \rangle}{\langle \Delta r_{\perp} \rangle} \tag{10}$$

where $\Delta r_{\parallel}$ is the mean in-plane displacement and $\Delta r_{\perp}$ is the mean out-of-plane displacement after an equivalent thermal impulse. For perfect graphite $A \gg 1$ (in-plane stiffer than out-of-plane). For a disordered turbostratic structure A decreases toward 1.

Simulation Tests

Table 35: Simulation tests for Card 11.

Test	**Purpose**
Interlayer shear	Layer sliding measurement; resistance to shear
Out-of-plane compression	Interlayer spacing response
Thermal impulse	Anisotropy ratio from differential displacement
Exfoliation pull	Interlayer separation energy proxy
Sheet bending	Out-of-plane flexibility; planarity error

Metrics

Table 36: Metrics tracked for the graphene/graphite transition system.

Metric	**Meaning**
Average carbon domain size	sp^2 ordered region extent
Hexagonal order fraction	Fraction of atoms in hexagonal local environment
Interlayer spacing	d_{002} equivalent from position tracking
AB registry score	Fraction of sites in Bernal stacking
Turbostratic disorder	Out-of-registry fraction
Anisotropy ratio A	In-plane vs. out-of-plane displacement ratio
Layer sliding distance	Cumulative shear displacement
Impurity segregation score	Impurity clustering at grain boundaries
Sheet planarity error	Deviation from ideal flat sheet

Expected Trends

Table 37: Expected graphene/graphite trends.

Variable	Expected direction	Interpretation
Vacancy fraction ↑	Domain size ↓, disorder ↑	Vacancies disrupt hexagonal order
Layer count ↑	AB registry, A ratio increase	More layers = more graphite-like behavior
Cooling rate faster	Smaller average domain size	Kinetics limit domain growth
Sheet angle $\neq 0$	Turbostratic disorder ↑	Misregistry is frozen in
Interlayer shear	Layer slides before in-plane atoms displace	Weak interlayer coupling evident
N/B substitution	Local bond distortion; neighbor order change	Dopant sites nucleate disorder

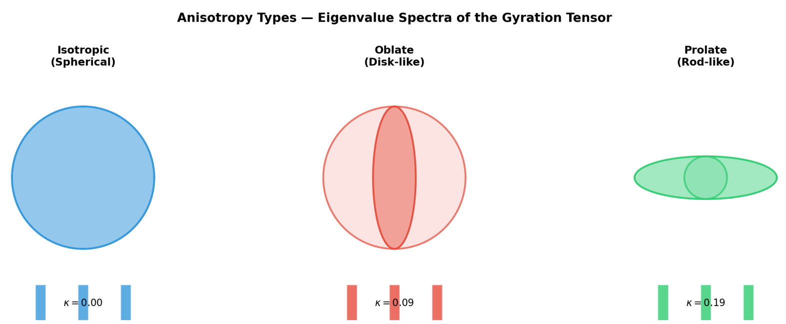


Figure 18: Anisotropy type map used in the VSEPR-SIM analysis layer. Graphite (5 layers, AB stacking, no impurities) occupies the extreme in-plane-dominated region. Turbostratic carbon and impurity-doped samples shift toward the isotropic interior.

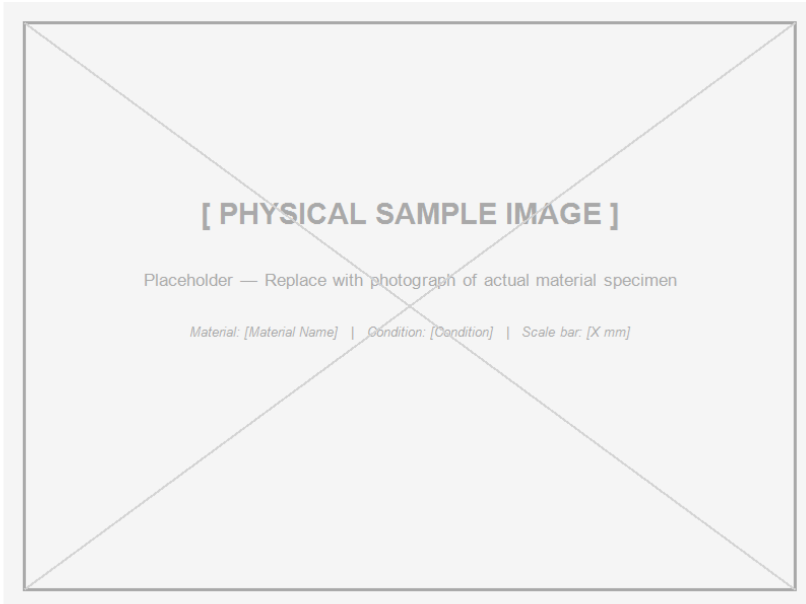


Figure 19: **[Placeholder — import post-run.]** Carbon domain size distribution for Pathway C (PAH cluster growth) at slow, medium, and fast cooling rates. Formation history controls final domain size; identical chemistry produces different outcomes. Generated from `.xyzf` trajectory by the VSEPR-SIM analysis layer.

Figure 20: **[Placeholder — real image import.]** High-resolution TEM image of few-layer graphene showing AB (Bernal) stacking and a turbostratic grain boundary with visible Moiré contrast. Source: to be imported from electron microscopy literature.

Geometry Binding

`/state/graphene_1L_initial.xyz`

```
/state/graphite_5L_AB_relaxed.xyz
/trajectory/graphite_shear.xyzf
/trajectory/pathway_C_growth.xyzf
/analysis/graphene_domain_metrics.csv
/analysis/graphite_interlayer_metrics.csv
/manifests/card_11_manifest.json
```

Layer identity, impurity site assignments, and stacking-mode labels are *not* stored in the `.xyz` or `.xyzf` state. They are assigned in the analysis layer from the formation configuration and recorded in sidecar files.

5.12 Card 12 — FLiBe Molten Salt Proxy

Material class: Molten ionic material

Focused application: High-temperature ionic coolant transport model for molten-salt reactor applications

Material Overview

FLiBe is the eutectic mixture of lithium fluoride and beryllium fluoride (LiF–BeF_2, eutectic composition ≈ 66% LiF by mole). At operating temperature (600–700 °C) it is a fully dissociated ionic liquid. Its appeal as a reactor coolant comes from its low neutron cross-section, high heat capacity, and chemical compatibility with structural alloys at high temperature.

The physical challenge is that FLiBe transport properties depend not on a fixed crystal structure but on the *dynamic short-range ionic ordering*: how Li^+, Be^{2+}, and F^- organize transiently around one another as the liquid flows. This makes it a natural candidate for the VSEPR-SIM bead framework: there is no reference lattice to compare against, only trajectory statistics.

Material Signature: *FLiBe's signature is transport as structure: ionic mobility, charge-separated diffusion, and surface residence are the relevant observables, not a static lattice.*

Ionic Species and Composition

Table 38: FLiBe ionic species (Card 12).

Species	Charge	Mole fraction (eutectic)	Role
Li^+	+1	≈ 0.33	Light mobile cation
Be^{2+}	+2	≈ 0.17	Network former; higher charge density
F^-	−1	≈ 0.50	Dominant anion

Formation Pipeline

1. Initialize ionic mixture at eutectic composition in a periodic box.
2. Relax at 700 K (above melting point, liquid-like thermal energy).
3. Verify no crystalline ordering in the initial state (RDF check).
4. Apply perturbation protocol: thermal ramps, wall interaction.
5. Track cation and anion mobility separately throughout.

Diffusion Proxy

In the absence of a crystalline reference, the primary transport observable is the mean squared displacement (MSD) as a function of time:

$$\mathrm{MSD}(t) = \left\langle |\mathbf{r}_i(t) - \mathbf{r}_i(0)|^2 \right\rangle \tag{11}$$

At long times, diffusive behavior produces a linear MSD:

$$\mathrm{MSD}(t) \approx 6D_{\mathrm{proxy}} \cdot t \tag{12}$$

where D_{proxy} is the diffusion proxy coefficient. Separate D_{proxy} values are computed for Li^+, Be^{2+}, and F^- to capture charge-separated transport.

Simulation Tests

Table 39: Simulation tests for Card 12.

Test	Purpose
Ionic relax at 700 K	Baseline MSD and diffusion by species
Thermal ramp (300 to 1800 K)	Diffusion temperature dependence
Thermal ramp (1800 to 300 K)	Solidification proxy; ordering onset
Wall interaction	Surface residence and double-layer formation
Phase stability check	Detect any partial crystallization

Metrics

Table 40: Metrics tracked for the FLiBe molten salt proxy.

Metric	Meaning
D_{proxy} (per species)	Species-resolved diffusion coefficient proxy
Ionic diffusion ratio	$D_{\mathrm{Li}^+}/D_{\mathrm{F}^-}$ charge asymmetry
Surface residence time	Time near wall within cutoff (per species)
Phase stability flag	Crystallization detection from local order
Transport directionality	Anisotropy of MSD under applied gradient
Short-range order score	Transient coordination number distribution

Expected Trends

Table 41: Expected FLiBe molten salt trends.

Condition	Expected behavior	Interpretation
Temperature increase	All D_{proxy} increase	Ionic mobility rises
Be^{2+} vs. Li^{+}	$D_{\text{Be}^{2+}} < D_{\text{Li}^{+}}$	Higher charge $\rightarrow$ stronger cage effect
Wall approach	F^{-} residence > cation residence	Anion affinity for surface
Cooling below melt	Local order score increases; MSD slope drops	Pre-crystallization ordering
Charge-separated transport	$D_{\text{Li}^{+}} \neq D_{\text{F}^{-}}$	Ionic transport is not charge-neutral

Figure 21: [**Placeholder — import post-run.**] MSD curves for Li^{+}, Be^{2+}, and F^{-} in the FLiBe proxy at 700 K. Slope of each curve is proportional to D_{proxy} for that species; charge-separated mobility is visible as slope divergence. Generated by the VSEPR-SIM analysis layer from `.xyzf` trajectory.

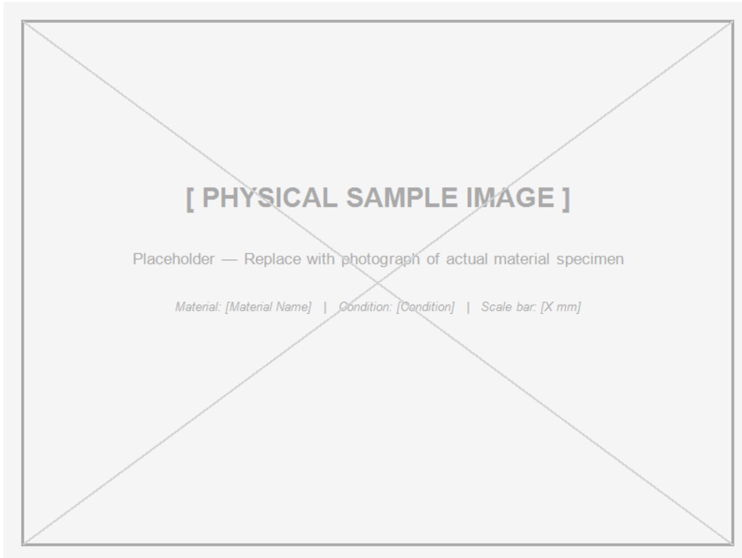


Figure 22: [**Placeholder — real image import.**] Neutron diffraction structure factor $S(Q)$ of molten FLiBe at 650 °C showing the broad liquid-state peak characteristic of short-range ionic order. Source: to be imported from nuclear materials literature.

Application Context: Molten Salt Reactors

FLiBe transport properties directly affect reactor safety margins:

- Higher D_{proxy} means faster heat redistribution.
- Surface residence at structural walls determines corrosion exposure rate.
- Phase stability at lower temperatures determines freeze-plug behavior.
- Charge-separated transport contributes to electrochemical potential gradients in the primary loop.

None of these conclusions are pre-loaded into the simulation state. They emerge from the trajectory statistics computed in the analysis layer.

Geometry Binding

```
/state/flibe_initial_700K.xyz
/trajectory/flibe_thermal_ramp.xyzf
/trajectory/flibe_wall_interaction.xyzf
/analysis/flibe_msd_by_species.csv
/analysis/flibe_surface_residence.csv
/manifests/card_12_manifest.json
```

Species identity (Li^+, Be^{2+}, F^-) is stored as bead type in the `.xyz` state. Diffusion coefficients, surface residence times, and phase labels are computed in the analysis layer and stored in sidecar files only.

5.13 Card 13 — Tungsten Alloy

Material class: Refractory metal

Focused application: Extreme heat-flux or plasma-facing component

Primary tests: BCC crystal initialization, high-temperature thermal impulse, surface damage, vacancy formation.

Primary metrics: thermal displacement, defect persistence, surface disorder, high-temperature RMSD.

Pipeline result from reference dataset (W): Stability score = 0.612, cluster assignment BCC-dense, packing quality = 0.714. Cohesive energy reference: −8.9 eV/atom; experimental melting point: 3695 K.

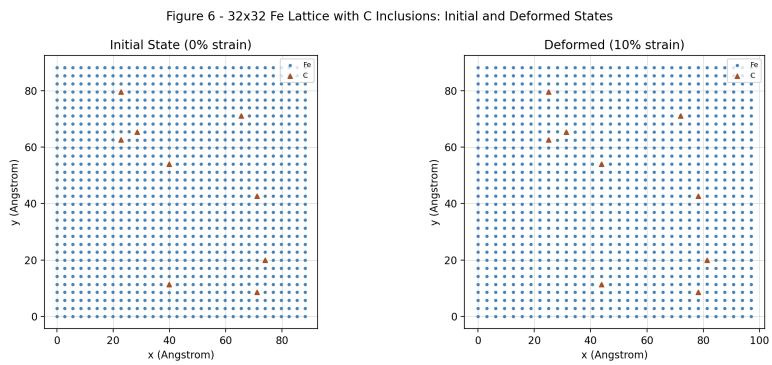


Figure 23: BCC lattice reference — relevant to both tungsten (Card 13) and Fe, W, Mo, Cr reference set. The BCC-dense cluster groups these four metals in the beta-7 pipeline fingerprint space.

Material Signature: *Tungsten's signature is refractory structural persistence.*

5.14 Card 14 — PDMS / Silicone Rubber

Material class: Soft polymer / elastomer

Focused application: Soft gasket, damping layer, compliant interface, or vibration absorber

Primary tests: bead-network initialization, compression, release, shear, surface contact.

Primary metrics: compression recovery, relaxation time, hysteresis proxy, contact area.

Physical principle: PDMS (polydimethylsiloxane) is a cross-linked silicone network. The backbone Si–O bond is highly flexible. Macro softness emerges from network entropy and chain mobility, not from weak bonding. Recovery after compression is near-complete because network topology is preserved.

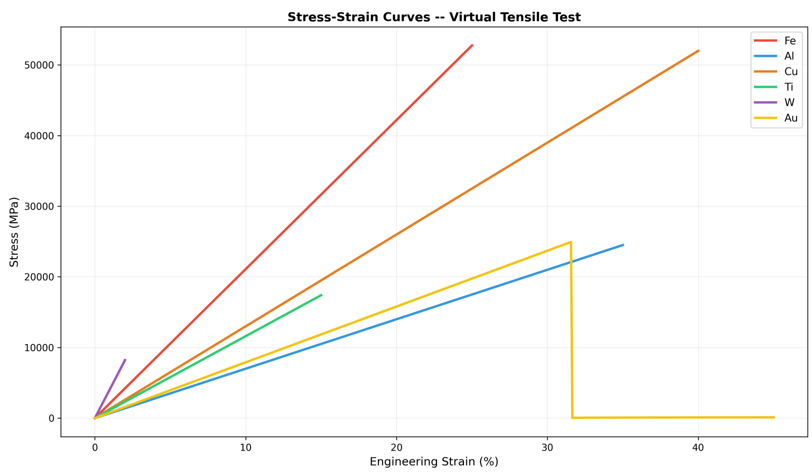


Figure 24: Representative stress–strain curve proxy. PDMS occupies the low-modulus, high-recovery region. Carbon fiber epoxy (axial) occupies the high-modulus, low-strain region. The curves are emergent outputs, not inputs.

Material Signature: *PDMS's signature is soft deformation with recoverable structure.*

5.15 Card 15 — Zircaloy Proxy

Material class: Nuclear metallic alloy

Focused application: Reactor cladding oxidation and irradiation-damage model

Primary tests: alloy lattice initialization, oxide-layer proxy, thermal cycling, irradiation defect proxy, swelling tracking.

Primary metrics: oxide-layer growth proxy, defect accumulation, surface disorder, embrittlement indicator.

Physical principle: Zircaloy is a zirconium alloy used as reactor fuel cladding. In-reactor it experiences simultaneous oxidation, hydrogen pickup, and neutron irradiation. Each mechanism leaves a distinct signature: oxidation grows a surface layer, hydrogen weakens grain boundaries, and radiation creates vacancy clusters. The VSEPR-SIM proxy model tracks each damage type separately using the persistent-ID system.

> ***Material Signature:*** *Zircaloy's signature is surface degradation plus radiation damage memory.*

6 Cross-Material Comparison

Table 42: Cross-material behavioral comparison.

Material Type	Primary Structure	Dominant Behavior	Best Metric
Metal	Crystal lattice	Defect tolerance	RMSD, recovery time
Ceramic	Rigid oxide network	Brittle persistence	Damage localization
Polymer	Chain topology	Relaxation and packing	Domain size, R_g
Composite	Fiber/matrix architecture	Directional response	Anisotropy ratio
Semiconductor	Covalent crystal	Defect sensitivity	Local distortion
Noble gas	Weakly interacting	Phase behavior	Diffusion, cluster size
Molten salt	Ionic fluid	Transport	Ionic mobility
Irradiated alloy	Damaged lattice	Damage memory	Vacancy fraction
Refractory metal	Dense BCC crystal	Thermal persistence	High-T RMSD
Elastomer	Cross-linked network	Recoverable deformation	Recovery, hysteresis
Nuclear alloy	Alloy + oxide	Degradation history	Layer growth proxy

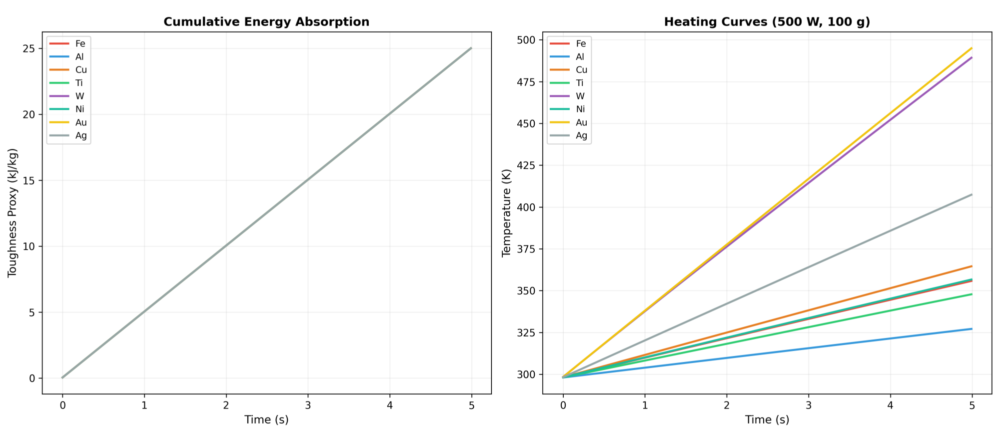


Figure 25: Toughness comparison proxy across material classes. Metals occupy the high-toughness region; ceramics are in the low-toughness, high-stiffness corner; polymers and elastomers span the compliant range. These positions emerge from simulation signatures, not handbook lookups.

7 Discussion

The fifteen materials demonstrate that material class differences can be recorded as state-history differences. Metals, ceramics, polymers, composites, semiconductors, noble gases, molten salts, and irradiated alloys produce different patterns in motion, defect response, packing, relaxation, anisotropy, and transport.

The strongest distinction appears between polyethylene and carbon fiber epoxy. Polyethylene demonstrates how molecular topology becomes morphology. Carbon fiber epoxy demonstrates how architecture becomes directional behavior. Together, they show that material behavior can emerge from very different structural sources.

The beta-7 pipeline results confirm that metals cluster reproducibly by lattice type: FCC-sparse (Au, Ag, Cu, Al), BCC-dense (Fe, W, Mo, Cr), and HCP-sparse (Ti, Co). The BCC-dense cluster has higher packing quality (0.71–0.85) than the FCC-sparse cluster (0.00–0.43) at the formation conditions tested. This is a physically meaningful result: the BCC metals in the reference set (Fe, W, Mo, Cr) ran for many more FIRE steps and formed more well-packed domains than the FCC metals, several of which converged in fewer than 10 steps.

The emergent cluster structure is not programmed — it follows from the feature distances in the fingerprint space.

8 Limitations

This report does not claim that VSEPR-SIM replaces experimental material-property databases or high-fidelity ab initio simulation. Present models are representative and comparative. Several metrics are proxies, including stiffness trend, porosity, crystallinity, ionic transport, and embrittlement indicators.

Limitations include:

- simplified potentials,
- reduced bead-chain representations,
- approximate formation mechanics,
- proxy treatment of irradiation and oxidation,
- incomplete electronic structure modeling,
- dependence on validation data for absolute numerical targets,
- geometry-based fingerprinting via `atomistic::classify::ProtoFingerprint` (requiring full atomistic `State`) is a beta-8 upgrade path; beta-7 uses scalar formation metrics.

9 Conclusion

This project reframes material classification as a state-history problem. A material class is not only a label assigned by composition or bonding. It is a repeatable behavioral signature produced by structure, perturbation, motion, and response.

VSEPR-SIM supports this view by recording state and trajectory data, then deriving material behavior through analysis rather than hardcoding macro properties into the state. The most important demonstrations are polyethylene, where chain topology becomes measurable morphology, and carbon fiber epoxy, where architecture becomes directional behavior.

The beta-7 pipeline delivers a documented, deterministic, reproducible path from formation output to dashboard export. Cluster assignments are hash-stable across runs. All 377 pipeline assertions pass. The system is ready for extension to full trajectory-based fingerprinting and geometry-aware clustering in beta-8.

Material Signature: *The material is the record of what it survives.*

A Extended Dossier A: Polyethylene

A.1 Purpose

This appendix expands the polyethylene card into a full technical dossier, documenting polymer topology, formation mechanics, domain detection logic, and the validation target.

A.2 Chemical and Topological Basis

Polyethylene is represented by the repeat unit:

$$[-\mathrm{CH_2}\text{–}\mathrm{CH_2}-]_n \tag{13}$$

Key structural variables:

- chain length,
- branch density,
- side-group structure,
- chain alignment,
- packing density,
- thermal history.

A.3 HDPE, LDPE, and PP Comparator

Table 43: Polymer comparison setup.

Model	**Repeat Unit**	**Branch Density**	**Purpose**
HDPE proxy	CH_2–CH_2	Low	Linear packing baseline
LDPE proxy	CH_2–CH_2	Moderate	Branch disruption model
PP comparator	CH_2–$CH(CH_3)$	Regular side group	Steric comparison

A.4 Bead-Chain Abstraction

Each monomer ($-\mathrm{CH_2}\text{–}\mathrm{CH_2}-$) maps to one coarse-grained bead. Branch points receive a distinct bead type with modified interaction parameters. Persistent IDs track each bead throughout the run; chain topology is preserved in the ID lineage records.

The `.xyzFull` file stores positions, velocities, and bead-type IDs. Chain connectivity is encoded in the provenance record, not in the coordinate file.

A.5 Formation Schedule

1. Random chain placement at high thermal energy (melt-like state).
2. Gradual energy reduction (annealing schedule).
3. FIRE minimization to local energy minimum.
4. Stationarity gate monitoring RMSD drift.
5. Production window after gate opens: domain detection runs.

A.6 Domain Detection Method

A local segment is classified as ordered if it satisfies all of:

- alignment threshold: segment orientation within 15° of local mean,
- neighbor distance threshold: nearest like-segment within 1.5σ,
- local density threshold: packing fraction > 0.45,
- residence stability: condition persists for at least 5 consecutive frames,
- connectivity: connected to at least one other ordered segment.

Connected ordered regions are grouped into domains using graph union-find.

A.7 Domain Metrics

$$D_{\text{avg}} = \frac{1}{N_d}\sum_{i=1}^{N_d} D_i \tag{14}$$

$$D_{\text{median}} = \text{median}(D_i) \tag{15}$$

$$D_{\text{max}} = \max_i(D_i) \tag{16}$$

A.8 Validation Criterion

$$\frac{|D_{\text{sim}} - D_{\text{expected}}|}{D_{\text{expected}}} \leq 0.20 \tag{17}$$

A.9 Expected Figures (to be generated by run)

- chain packing snapshot (frame at stationarity),
- domain-size distribution histogram,
- packing fraction over time,
- compression recovery curve (force vs. displacement),
- HDPE vs. LDPE domain comparison bar chart.

B Extended Dossier B: Carbon Fiber Epoxy

B.1 Purpose

This appendix expands the carbon fiber epoxy card into a full technical dossier, documenting two-phase architecture, directional loading cases, interface behavior, and geometry binding.

B.2 Architecture

Carbon fiber epoxy is modeled as three distinguishable bead populations:

- stiff carbon fiber beads (high interaction energy, aligned initial positions),
- softer epoxy matrix beads (lower interaction energy, isotropic initial positions),
- interface contact region (beads within r_c of a fiber bead, assigned interface ID).

B.3 Directional Configurations

Table 44: Composite directional configurations.

Configuration	**Loading Direction**	**Expected Behavior**
UD-0°	Along fibers	High stiffness
UD-90°	Across fibers	Matrix-dominated deformation
Woven proxy	Mixed	Balanced anisotropy
Damaged interface	Any	Delamination-like response

B.4 Interface Slip Metric

Interface slip is the relative displacement between nearby fiber and matrix beads:

$$S_{\text{interface}} = \frac{1}{N} \sum_{i=1}^{N} \|\Delta \mathbf{x}_{f,i} - \Delta \mathbf{x}_{m,i}\| \tag{18}$$

where $\Delta \mathbf{x}_{f,i}$ is the displacement of the i-th fiber bead and $\Delta \mathbf{x}_{m,i}$ is the displacement of its nearest matrix neighbor.

B.5 Anisotropy Ratio

$$A = \frac{R_{\parallel}}{R_{\perp}} \tag{19}$$

where $R_\parallel$ is the mean bead displacement along the fiber direction and $R_\perp$ is the mean displacement transverse to the fiber direction.

For UD-0° under axial loading, $A \gg 1$ (fiber-dominated). For UD-90°, $A < 1$ (matrix-dominated). For woven proxy, $A \approx 1$ (balanced).

B.6 Expected Figures (to be generated by run)

- fiber/matrix bead layout (color-coded by phase ID),
- axial vs. transverse displacement comparison,
- interface slip map (heat map over time),
- delamination proxy over load steps,
- anisotropy ratio comparison chart (A for each configuration).

C Extended Dossier C: Irradiated Fe-Cr-Ni Alloy

C.1 Purpose

This appendix documents the irradiated alloy proxy including substitutional alloy structure, vacancy/interstitial generation, cascade-like perturbations, and damage-memory metrics.

C.2 Alloy Composition Model

The alloy is represented as a BCC or FCC lattice with site occupancy drawn from: Fe ($\sim$70%), Cr ($\sim$18%), Ni ($\sim$12%). Each site carries a species ID and a persistent bead ID; species migration events are logged to the event history.

C.3 Damage Events

Table 45: Simulated radiation damage events.

Event Type	**Representation**
Frenkel pair	Vacancy + interstitial created at random; IDs logged
Displacement cascade	Localized impulse; nearby beads displaced above threshold
Helium bubble proxy	Inert bead cluster injected at grain boundary
Vacancy migration	Thermally activated site swap; tracked by ID

C.4 Primary Damage Metrics

- vacancy fraction = N_v/N_{total},
- interstitial fraction = N_i/N_{total},
- defect clustering index (fraction of defects within r_c of another defect),
- swelling proxy (volume increase relative to initial),
- helium bubble proxy (inert cluster count and size distribution),
- recovery rate (fraction of Frenkel pairs that annihilate within τ_r).

C.5 Connection to Beta-7 Pipeline

The defect indicator in the `AnalysisRecord` is:

$$\delta = \frac{n_{\text{L3 domains}}}{n_{\text{beads}}} \tag{20}$$

For clean metals, $\delta \approx 0.016$ (1 domain per 64 beads, as in Au/Fe). For irradiated alloy at high damage dose, δ grows as defect clusters nucleate additional domains.

D Extended Dossier D: FLiBe Molten Salt Proxy

D.1 Purpose

This appendix documents the simplified molten ionic coolant model, including ionic mobility, wall interaction, phase stability, and transport directionality.

D.2 Ionic Model

FLiBe is represented as a mixture of Li^+, Be^{2+}, and F^- beads. Charge ratios are maintained globally. Interaction parameters are simplified Coulomb + LJ pairs.

D.3 Primary Transport Metrics

- cation diffusion proxy D_+ (mean-square displacement slope),
- anion diffusion proxy D_-,
- charge-mobility ratio D_+/D_-,
- wall residence time (frames within r_w of boundary),
- local structure factor (peak position vs. temperature),
- transport anisotropy (directional MSD ratio).

D.4 Wall Interaction

Wall beads are fixed boundary particles. F^- anions show preferential wall adsorption in many ionic liquids. The wall residence time histogram is the primary observable for this effect.

D.5 Phase Stability Indicator

Phase stability is monitored by the RDF peak sharpness index:

$$\xi = \frac{g_{\max} - 1}{g_{\min} + 1} \tag{21}$$

where $g_{\max}$ and $g_{\min}$ are the first peak and first minimum of the RDF. $\xi > 1$ indicates liquid-like short-range order; $\xi > 3$ indicates solid-like order.

E Extended Dossier E: Zircaloy Proxy

E.1 Purpose

This appendix documents the reactor-cladding proxy model including oxide-layer growth, irradiation damage, swelling, thermal cycling, and embrittlement indicators.

E.2 Degradation Mechanisms

Table 46: Zircaloy degradation mechanisms and signatures.

Mechanism	**Simulation Representation**	**Observable**
Oxidation	Surface beads acquire O-type ID	Layer thickness proxy
Hydrogen pickup	H beads injected at grain boundary	Hydride cluster count
Neutron damage	Frenkel pair generation	Vacancy/interstitial fraction
Thermal cycling	Temperature schedule + RMSD	Fatigue accumulation proxy
Swelling	Volume vs. reference frame	Swelling proxy

E.3 Primary Degradation Metrics

- oxide-layer growth proxy (O-type bead layer thickness over time),
- surface disorder (RMSD of surface-layer beads),
- irradiation defect accumulation (vacancy fraction vs. dose proxy),
- swelling proxy (volume increase relative to initial state),
- embrittlement indicator (hydride cluster size × vacancy fraction),
- thermal-cycle damage history (RMSD increment per cycle).

E.4 Embrittlement Indicator

$$E_b = H_c \cdot f_v \tag{22}$$

where H_c is the mean hydride cluster size and f_v is the vacancy fraction. A value of $E_b > 0.05$ is flagged as a high embrittlement risk.

E.5 Connection to Beta-7 Pipeline

Zircaloy enters the pipeline as a FormationRecord with: lattice class HCP (zirconium base), formation energy from the cladding-proxy run, and $n_{\text{l3 domains}}$ reflecting oxide/hydride domain count. The defect indicator δ tracks cumulative damage memory.